



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

Nuclear Starburst Disks

ACTIVE GALACTIC NUCLEI

Fritz Ali Agildere

April 29, 2026

Motivation

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- Natural Central SMBH Rapid Feeding Consequence

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- Fuel Region / Galactic Disk Kinematic Decoupling

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- Dynamically Evolving Obscuring Starforming Structure



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- Highly Localized $R < 1 \text{ pc}$ Obscuration (Compton Thick)

Structure

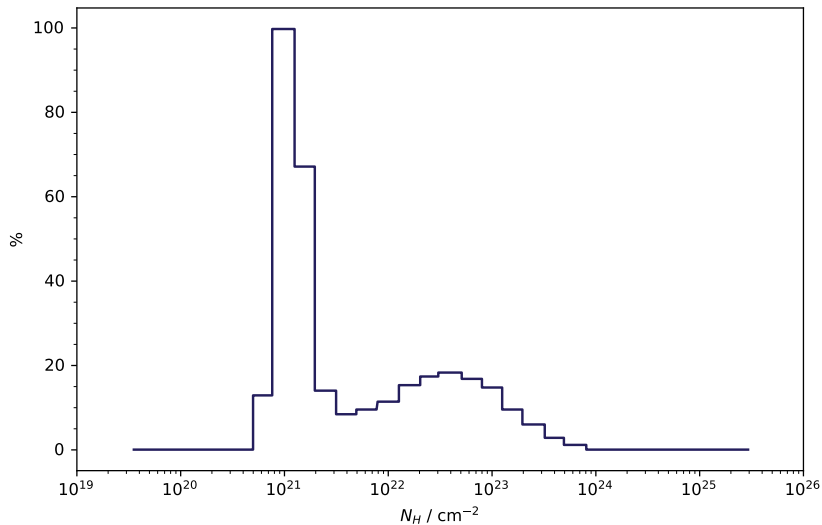
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- Statistically Consistent Viewing Angle Sampled Distributions

Column Density

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Modeling



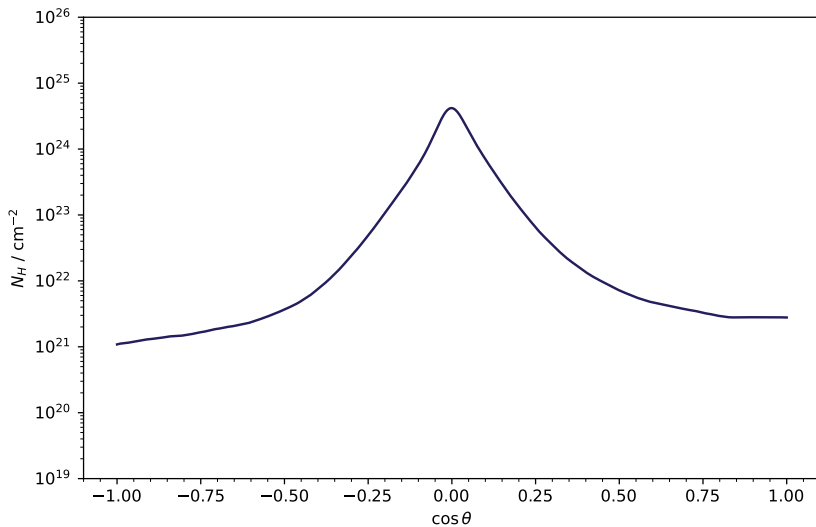
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Angular Obscuring

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Outlook

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- CO SLED Molecular Gas Tracers (ALMA)
- IR Starforming & Disk Signatures (JWST)

Thank You!

- R. C. Hickox & D. M. Alexander, *Obscured Active Galactic Nuclei* (2018) ARA&A Volume 56
- P. F. Hopkins et. al., *Stellar and Quasar Feedback in Concert: Effects on AGN Accretion, Obscuration and Outflows* (2016) MNRAS Volume 458 Issue 1

Resources on GitHub: [here](#)